

Disease Prediction System

Abstract

Disease Prediction System is a machine learning project used to predict diseases from patient clinical data. The system can be used for diseases such as diabetes and heart disease. It takes information such as age, blood pressure, glucose level, cholesterol and other medical values as input and predicts whether the patient may have the disease.

In this project, four machine learning algorithms are compared: Decision Tree, Random Forest, KNN and Naïve Bayes. The data is first cleaned and divided into training and testing data. Each algorithm is trained and tested using the same dataset. Their performance is compared using accuracy, precision, recall, F1-score and confusion matrix.

The main aim of this project is to find which algorithm gives better results for disease prediction. This project helps students understand how different machine learning algorithms can be applied to healthcare data. The system is for learning and prediction purposes and does not replace a doctor's diagnosis.

Datasets

Dataset	Source
Diabetes Dataset	Pima Indians Diabetes Dataset
Heart Disease Dataset	UCI Heart Disease Dataset

Approach

- 1. Collect Data:** Take a diabetes or heart disease clinical dataset.
- 2. Preprocess Data:** Clean the data and prepare the required features.
- 3. Split Data:** Divide the dataset into training and testing data.
- 4. Apply Models:** Train Decision Tree, Random Forest, KNN and Naïve Bayes.
- 5. Predict Disease:** Test the models using patient clinical data.
- 6. Compare Models:** Compare accuracy, precision, recall and F1-score to find the better model.

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